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# Software Development Manual

## 一、Basic information

### 1. Factory Default Configuration

Communication Mode：USB

Trigger Mode：Manual mode

Terminator：CR。



Factory Default Configuration

### 2. Save current configuration as default

The user can set the required configuration , and then scan the following barcode，Save current configuration as default



Save current configuration as default

### 3. Default Configuration

When the user has set the default settings, scan the following barcode to restore the original set of customer configurations



Default Configuration

**4. Parameter code**



***open Parameter code**



**close Parameter code**

**5. Product information**



## 二、Communication Mode

**1.** Serial Port

Using Serial Port，read moudle and the host devices must match exactly in ommunication parameter configuration, to ensure smooth communication and content are correct, Serial Port is configured to: **9600 baud, 8 bits of data, no parity, 1 stop bit**



**TTL 232**

（1）Baud Rate

The default baud rate is 9,600



**1200bps**



**2400bps**



4800bps



* 9600bps



19200bps



38400bps



57600bps



115200bps

（2）parity



Odd



Even



* None

（3）Stop bit



***1 Stop Bit**



**2 Stop Bits**

**2. USB KBW**



**USB KBW**

**3. USB COM**



**USB COM**

## 三、Trigger Mode

### 1. Power Mode

This parameter determines the power mode of the engine.

In Low Power mode, the scan engine enters into a low power consumption Sleep power state whenever possible (provided all WAKEUP commands were released)

In Continuous Power mode, the scan engine remains in the Awake state after each decode attempt

The Sleep and Awake commandscan be used to change the power state in either the Low Power mode or the Continuous Power mode.



**Continuous Power**



**Low Power**

### 2. Manual mode

**（1）Key Holding**

Press the button to trigger the reading, release the button to end the reading. Reading success or reading time over a single reading time will end the reading。



***Manual mode-Key Holding**

**（2）Single Key Trigger**

Detects the change of the key level (Maintain 30ms, depending on the product )to start reading, and then detects the change of the key level (Maintain 30ms, depending on the product )again to end reading. Reading success or reading time over a single reading time will end the reading.



**Manual mode-Single Key Trigger**

### 3. Continuous Mode

The reading engine performs continuous work. Reading success or reading time over a single reading time will end the reading. More than the specified time will automatically trigger the next reading



Continuous Mode

#### （1）Interval Time

The interval time between two readings in continuous mode. Regardless of the last success or failure to read, more than the specified time will automatically trigger the next reading.

Default: 500ms,unit: 100ms,range: 0-9900ms

To set a Interval Time, scan the bar code below. Next scan two Numeric Bar Codes in appendix that correspond to the desired time-out. Single digit values must have a leading zero. For example, to set a time-out of 0.5 seconds, scan the bar code below, then scan the “0” and “5” bar codes. To change the selection or cancel an incorrect entry, scan Cancel



**Interval Time**

**(Default: 500ms.)**

## 4. Automatic Induction Mode

In automatic induction mode, the scan engine detects the brightness of the surroundings. Trigger reading when the brightness changes. Reading success or reading time over a single reading time will end the reading. Regardless of the last success or failure to read, re-enter the detection of the surrounding environment brightness.



**Automatic Induction Mode**

### （1）Stability of Induction Time

Stability of induction time, Default: 500ms, unit:100ms, range: 0-9900ms

For example:

Set stability of induction time is 200ms

Scan stability of induction time setting code,then scan *Numeric Bar Codes* 0 and 2

Set stability of induction time is 1500ms

Scan stability of induction time setting code,then scan *Numeric Bar Codes* 1 and 5



**Stability of Induction Time**

### （2）Sensitivity Level

There are three levels of sensitivity to choose from , Default: 500ms



***High**



Middle



**Low**

**5. Host mode**

Through the command to trigger the scan engine to read, also through the command to trigger the scan engine to end reading. Reading success or reading time over a single reading time will end the reading.



Host mode

**6. Duration in Scanning**

This parameter sets the maximum time decode processing continues during a scan attempt. It is programmable in 0.1 second increments from 0.50 to 25.5 seconds.

To set a duration in scanning, scan the bar code below. Next scan three *Numeric Bar Codes* in appendix that correspond to the desired on time. Single digit numbers must have a leading zero. For example, to set an

on time of 0.5 seconds, scan the bar code below, then scan the "0", "0" and "5" bar codes; to set an on time of 10.5 seconds, scan the bar code below, then scan the "1", "0" and "5" bar codes. To change the selection or cancel an incorrect entry, scan *Cancel*



**Duration in Scanning(Default: 3.0 sec.)**

### 7. Output Interval of The Same Code

To avoid reading the same barcode multiple times in continuous mode and automatic induction mode, set the scan engine to allow reading the same barcode after a delay.

Output interval of the same code is to refuse to read the same barcode within the set length of time.

Default: 500ms,unit:100ms,range: 0-9900ms

To set output interval of the same code, scan the bar code below. Next scan two Numeric Bar Codes in appendix that correspond to the desired time-out. Single digit values must have a leading zero. For example, to set a time-out of 0.5 seconds, scan the bar code below, then scan the “0” and “5” bar codes. To change the selection or cancel an incorrect entry, scan Cancel in appendix.



**Output Interval of The Same Code**

### 8. Quick set for output Interval of The Same Code



None



delay 1s



delay 3s

Delay 5s

delay 7s

delay forever

## 四、Floodlight and Positioning lights

### 1. Floodlight

*** Lighting when Read**

Always Lighting

**Always Close**

### 2. Positioning lights

*** Lighting when Read**



Always Lighting



**Always Close**

## 五、Output and prompt

### 1. Keyboard

#### （1） Country/Language Keyboard



* American Keyboard



**Belgium**



Finland



**France**



Germany



**Italy**



Sweden



**England**



Denmark



**Norway**



Spain



**Portugal**



Turkey_F



**Turkey_Q**



Japan



Russia



Czech



Thailand



Ukraine



Brazil（ABNT2）



Greece



Hungary



Netherlands



Poland（214）



Romania (standard)



Slovakia



Multi-national keyboard

（2） **Keyboard type**

Enable virtual keyboard, you can output the correct data in any keyboard language mode.When using virtual keyboard, you must ensure that the keypad keys are valid。



*StandardKeyboard



Virtual Keyboard

**（3） Time interval that keyboard outputs character**

Time interval that keyboard outputs character, range: 0-1000ms,unit: 5ms,default: 5ms



0ms



10ms

**（4） Control characters in ASCII output mode select**

Control characters in ASCII (0x00-0x20) output mode select.

**output function keys**: control characters are used as custom function keys, the function is detailed in appendix 4

**output Ctrl key combination (this function is used with the prefix)** : CTRL key combination mode output control character, detailed function is detailed in appendix 4

**ALT mode output control character:** support full control character output in Chinese environment, specific Reference Standard ASCII table

**output Enter, DownArrow**: Shielding other control characters, only output: 0x07 output Enter, 0x0A output DownArrow, 0x0D output Enter.



**output function keys**

output Ctrl key combination

ALT mode output control character

Output Enter、DownArrow

2. **Prompt sound**

（1） **type of Buzzer**

*Passive Buzzer

Active Buzzer

（2） **Mute**

Open

* Close

**（3） Beeper Volume**



* High



**Middle**



Low

**（4） Beep After Good Decode**



*Open



**Close**

**（5） Boot prompt**



*open



**Close**

**（6） Setup Code Prompt**



*open



**Close**

**3. LED After Good Decode**

To enable or disable LED after good decode, scan the appropriate bar code below



**Disable**



***enable**

**4. operation Mode of decode led**



***the indicator led is always on**



***the indicator led is always off**

**5. Transmit “No Read” Message**

Enable this option to transmit “NR” if a symbol does not decode during the timeout period or before the trigger is released. Any enabled prefix or suffixes are appended around this message.

When disabled, and a symbol cannot be decoded, no message is sent to the host.



***Disable No Read**



**Enable No Read**

**6. Letter case conversion**

For example If the Barcode content is： ab123dE，if set to " all uppercase "，the output is：AB123DE；if set to "all lowercase", the output is：ab123de；

if set to " Case Inversion", the output is：AB123De；

Default: **Normal Letter Case**



* Normal Letter Case



all uppercase



all lowercase



Case Inversion

**7. Output Character Set Type**

0: Primitive Type

1:GBK(GB2312)

2: UTF8



Primitive Type



***GBK**



**Unicode**

**8. Input Character Set Type**





*AUTO

GBK(GB2312)



UTF8



ASCII



Japanese



DEC MULTI-COUNTRY CHARACTER SET (MCS)



Japanese single byte

## 六、Data editor

### 1. Code ID

The user can identify different barcode types by CODE ID, and CODE ID USES a character to identify them,detail in Appendix 3。



*Disable send Code ID



Enable send Code ID

**2. Terminator**

Add character format: Decode Data+Terminator.



***NONE**



**CR LF**



**CR**



**TAB**



**CRCR**

CR LF CR LF

**3. Terminator conversion**

LF(\n) and CR LF (\r\n) both convert CR (\r)

*disable

able

**4. Add multiple Prefixes/suffixes**

- Prefixes

(1) Scan following barcode“set multiple prefixes”

**set multiple prefixes**

(2) Next scan four Numeric Bar Codes in appendix,Scan the Numeric Bar Codes in turn, and set the successful prompt once every four times

(3) Scan following barcode“Complete setup multiple Prefixes/suffixes”

**Complete setup multiple Prefixes/suffixes**

- **suffixes**

（1）Scan following barcode“set multiple suffixes”



**set multiple Suffix**

（2）Next scan four Numeric Bar Codes in appendix,Scan the Numeric Bar Codes in turn, and set the successful prompt once every four times

（3）Scan following barcode“Complete setup multiple Prefixes/suffixes”



**Complete setup multiple Prefixes/suffixes**

- **Prefixes/suffixes take effect**



***Output Decoding Data Only**



**Prefixes+data**



**Data+suffixes**



**Prefixes+data+suffixes**

**5. Add Prefix and suffix base on bar code type (non-generic version)**

**(1) Add Prefix**



**Sets the prefix based on the bar code type**

Set multiple prefixes based on the bar code type,

**step 1:** Scan the Code First,

**step 2:** Select the Bar Code type, according to the bar code type supported by "Appendix 5" and "Appendix 4", scan the index of the corresponding hexadecimal value, such as QR Code type 0XF1, 0XF1 Hexadecimal to decimal is 241, the scan valve is 1000+241=1241, then scan the 1,2,4,1 Numeric Bar Codes to select the type of Bar Code;

**Step 3:** Scan the required prefix, such as the need to set the number "1" , scan 1049(0x31) ;

**Step 4:** Scan "complete the setting of multiple prefix"



complete the setting of multiple prefix

**(2) Add Suffix**

Step refers to "add prefix bar code by type. "



**Sets the suffix based on the bar code type**

**(3) remove prefixes based on Bar code type**



remove prefixes based on Bar code type

**step 1:** Scan the Code First;

**step 2:** Select the Bar Code type, according to the bar code type supported by "Appendix 5" and "Appendix 4", scan the index of the corresponding hexadecimal value, such as QR Code type 0XF1, 0XF1 Hexadecimal to decimal is 241, the scan valve is 1000+241=1241, then scan the 1,2,4,1 Numeric Bar Codes to select the type of Bar Code;

**Notes:** if you want remove all type of barcode, Scan "1, 2, 5, 5" (0xFF)

**(4) remove suffixes based on Bar code type**

Step refers to "remove prefix bar code by type. "



Remove suffixes based on Bar code type

**(5) prefix and suffixes Switch setting（Data format）**



**No prefix and suffix (raw data)**



**Output prefix (prefix+data)**



**Output suffix (data + suffix)**



**Output prefix and suffix (prefix + data + suffix)**

**6. Hide data**

**(1) Hide Head Data**



***Disable**



**enable**

**Set Hidden Number**

range 1-255. Scan the following barcode,Next scan three Numeric Bar Codes in appendix. For example, if you need to hide 16 characters, scan three Numeric Bar Codes in turn: 0 1 6



Hide Head Data-head

**(2) Hide intermediate data**



***Disable**



**enable**

**Sets the start position of hidden intermediate data**

Sets the start position of hidden intermediate data, range 1-255.Scan the following barcode,Next scan three Numeric Bar Codes in appendix, for example, to hide the data after the third character(the fourth begins to hide),scan three Numeric Bar Codes in turn: 0 0 3



start position of hidden intermediate data

**Set Hidden Number**

range 1-255. Scan the following barcode,Next scan three Numeric Bar Codes in appendix. For example, if you need to hide 16 characters, scan three Numeric Bar Codes in turn: 0 1 6



Set Hidden Number-intermediate

**(3) Hide tail data**

*Disable

enable

**Set Hidden Number**

range 1-255. Scan the following barcode,Next scan three Numeric Bar Codes in appendix. For example, if you need to hide 16 characters, scan three Numeric Bar Codes in turn: 0 1 6



Set Hidden Number-tail

**7. Hide header and footer data base on bar code type (non-generic version)**

**（1）Hide header data**

**step 1**: Scan the Code First;

**step 2**: Select the Bar Code type, according to the bar code type supported by "Appendix 5" and "Appendix 4", scan the index of the corresponding hexadecimal value, such as QR Code type 0XF1, 0XF1 Hexadecimal to decimal is 241，the scan valve is 1000+241=1241，then scan the 1,2,4,1 Numeric Bar Codes to select the type of Bar Code;

**Step 3**: Hidden Length (4-digit decimal value).if you want hide 5 bit header data,scan 0,0,0,5 Numeric Bar Codes;

**Notes:** Supports setting all bar code types at once , Scan“1，2，5，5”（0xFF）



**Hide header data base on barcode type**

**（2）able/disable hide header data**



***disable hide header data**



**able hide header data**

**（3）Hide footer data base on bar code type**

Step refers to " **Hide header data base on barcode type**"



Hide footer data base on bar code type

**（4）able/disable hide footer data**



***disable hide footer data**



**able hide footer data**

## 8. Insert customize data

### （1）able /disable insert customize data

Supports customize data insertion anywhere in the Barcode, up to 10 bytes



***disable**



**able**

### （2）set the location where the data is inserted

Depending on where you want to insert, scan the corresponding value (4 digit Numeric Bar Codes) , such as set after the third character, you need to scan four Numeric Bar Codes "0" , "0" , "0" , "3" .

If set to 0, the header is inserted into the decoded data. If the setting is larger than the length of the decoded data, the end of the decoded data is inserted by default. The range of supported insertion positions is 0-5000.



set the location where the data is inserted

**（3）set the inserted data**

Set Insert customize data, such as need to set the character "QR" (0x51,0X52) , then scan two consecutive sets of Numeric Bar Codes 1081(1000 + 0x51) , 1082(1000 + 0x52) .



set the inserted data

**（4）exit settings customize data**

Maximum support for 10 bitye customize data, can be set continuously, over 10 bitye automatic exit settings. Complete ahead of time: scan the "exit settings customize data " setting code, exit settings and save the current set of data.



exit settings customize data

## 9. STX&ETX



*Disable



STX



ETX



STX+ETX

## 10. Character Replace

The feature can replace any character in Barcode data with another character, and supports either 1:1 or 1: n replacement, such as A--B、A--BC、A--BCD

Operating instructions:

**（1）Sets the data to be replaced**



Sets the data to be replaced

For example:Group Separator, According to the Character Equivalents,detali in Appendix 4,the scan value is 1029,scan the Numeric Bar Code1,0,2,9 one by one

**（2）Set the Replacement data**



Set the Replacement data

For example: symbol }， According to the Character Equivalents,detali in Appendix 4,the scan value is 1125,scan the Numeric Bar Code1,1,2,5 one by one

**（3）complete setup**



**complete setup**

**（4）Enable / disable data replacement**



**Enable**



**disable**

## 七、Code Enable/Disable

1. All Barcode Switch



enable



disable

2. 1d code master switch



enable



*disable

3. 2d code master switch



enable



*disable

4. Reverse code reading

Mainly for the 1D code switch, the 2D code is recommended to open separately,2D code has a separate switch



enable



*disable

5. Read Multiple Barcodes at once

（1） Must read Multiple Barcodes at once



able



disable

**（2） number of barcodes read at one time**



**1**



**2**



**3**

## 6. UPC-A

### （3） UPC-A enable/disable



*enable



disable

### （4） UPC-A Preamble



No Preamble



*System Character



System Character & Country Code

### （5） UPC-A check bit



Do not transmit UPC-A check bit



*transmit UPC-A check bit

## 7. UPC-A additional code

### (1) UPC-A 2 additional code



enable



***disable**

**(2) UPC-A 5 additional code**



Enable



*disable

**(3) UPC-A Must read additional code**



Enable



*disable

## 8. UPC-E

**（1） UPC-E enable/disable**



*****enable



disable

**（2） UPC-E Preamble**



No Preamble



*System Character



**（3） UPC-E check bit**



Do not transmit UPC-E check bit



*transmit UPC-E check bit

## 9. UPC-E additional code

**(1) UPC-E 2 additional code**



**enable**



***disable**

**(2) UPC-E 5 additional code**



**Enable**



***disable**

**(3) UPC-E Must read additional code**



**Enable**



***disable**

## 10. UPC-E transfer UPC-A



Enable



*disable

### 11. UPC-A transfer EAN-13



Enable



*disable

### 12. UPC-E1



enable



*disable

### 13. EAN-8

**（1） EAN-8 able/disable**



*enable



disable

**（2） EAN-8 check bit**



Do not transmit EAN-8 check bit



transmit EAN-8 check bit

### 14. EAN-8 additional code

**(1) EAN-8 2 additional code**



Enable



*disable

**(2) EAN-8 5 additional code**



Enable



***disable**

**(3) EAN-8 Must read additional code**



Enable



*disable

**15. EAN-13**

**（1） EAN-13 enable/disable**



***enable**



*disable

**（2） EAN-13 check bit**



Do not transmit EAN-13 check bit



*transmit EAN-13 check bit

**16. Bookland EAN (ISBN)**

ISBN disable is treated as EAN13

able ISBN

*disable ISBN

### 17. ISSN

ISSN disable is treated as EAN13

able ISSN

*disable ISSN

### 18. EAN-13 additional code

**(1) EAN-13 2 additional code**

Enable

*disable

**(2) EAN-13 5 additional code**

enable

*disable

**(3) EAN-13 Must read additional code**

Enable

*disable

## 19. CODE 128



* Enable



Disable

## 20. GS1-128



* enable



**disable**

## 21. Interleaved 2 of 5

**(1) I 2 of 5 enable/disable**



*enable



disable

**(2) Set Lengths for Interleaved 2 of 5**

For example, to decode **Interleaved 2 of 5** symbols containing between 4 and 12 characters

first scan **Interleaved 2 of 5-Length Within Range**, then scan **0**, **4**, **1** and **2** (single digit numbers must be preceded by a leading zero). *Numeric Bar Codes* is in appendix. To change the selection or cancel an incorrect entry, scan Cancel



**I 2 of 5 - Length Within Range**



**I 2 of 5 - Any Length**

**(3) I 2 of 5 Check Digit Verification**



enable



*disable

**(4) Transmit I 2 of 5 Check Digit**



enable



*disable

## 22. Matrix 2 of 5

**(1) Matrix 2 of 5 enable/disable**



enable



*disable

**(2) Set Lengths for Matrix 25**

For example, to decode Matrix 25 symbols containing between 4 and 12 characters

first scan **Matrix 25 Length Within Range**, then scan **0**, **4**, **1** and **2** (single digit numbers must be preceded by a leading zero). *Numeric Bar Codes* is in appendix. To change the selection or cancel an incorrect entry, scan *Cancel* in appendix.



**Matrix 25 - Length Within Range**



**Matrix 25 - Any Length**

**(3) Matrix 25 Check bit Verification**



Enable



*disable

**(4) transmit Matrix 2 of 5 check bit**



Enable



*disable

## 23. Industrial 2 of 5

**(1) Industrial 2 of 5 enable/disable**



**enable**



***disable**

**(2) Set Lengths for Industrial 2 of 5**

For example, to decode **Industrial 2 of 5** containing between 4 and 12 characters

first scan **Industrial 2 of 5 Length Within Range**, then scan **0**, **4**, **1** and **2** (single digit numbers must be preceded by a leading zero). Numeric Bar Codes is in appendix. To change the selection or cancel an incorrect entry, scan Cancel



**D 2 of 5 - Length Within Range**



**D 2 of 5 - Any Length**

## 24. Standard 2 of 5

**(1) Standard 2 of 5 enable/disable**



**enable**



***disable**

**(2) Set Lengths for Standard 2 of 5**

For example, to decode **Standard 2 of 5** containing between 4 and 12 characters

first scan **Standard 2 of 5 Length Within Range**, then scan **0**, **4**, **1** and **2** (single digit numbers must be preceded by a leading zero). Numeric Bar Codes is in appendix. To change the selection or cancel an incorrect entry, scan Cancel



**Standard 25 - Length Within Range**



**Standard 25 - Any Length**

## 25. Code 39

**(1) code39 enable/disable**



***enable**



**disable**

**(2) Code39 Length**



Any Length code39

**(3) Code 39 Check bit Verification**



enable



*disable

**(4) transmit Code39 check bit**

Transmit the check bit, the check bit Verification function should enabled



transmit



*do not transmit

**(5) Transmit Code 39 start and ending symbol**



*disable



enable

## 26. Code 39 Full ASCII



Enable

* Disable

## 27. Code 32

**(1) code32 enable/disable**



Enable

* Disable

**(2) code32 add prefix A**



Enable



* disable

## 28. Code 93

**(1) Code93 enable/disable**



**Enable**



*** disable**

**(2) code93 lenght**



**Any lenght can read**

## 29. Code 11

**（1） code11 enable/disable**



Enable



* disable

**（2） code11 lenght**



Any lenght can read

**（3） Code 11 Check bit Verification**



enable



**one check bit**



**two check bit**

**（4） transmit check bit**



enable



* disable

## 30. Codabar

**（1） Codabar enable/disable**



enable



* disable

（2） **Codabar lenght**



Any lenght can read

（3） **Start Character and Terminator**

The start character and terminator are allowed to be one of the four characters of “A”, “B” “C” “D”.The terminator is also allowed to be one of the four characters of “T”, “N”, “*”, “E”.



***ABCD/ABCD**



**ABCD/TN*E**

（4） **Transmit Start Character and Terminator**



Don not transmit



*transmit

**31. MSI**

（1） **MSI enable/disable**



Enable



*disable

（2） **MSI lenght**



Any lenght can read

### 32. GS1-Databar



**Enable**



*** disable**

### 33. GS1 composite code



Enable



*** disable**

### 34. QR Code

**(1) QR code enable/disable**



*** enable**



**disable**

**(2) QR code reverse**



*only read Normal code



**Read** Normal/reverse code

**(3) QR Mirror**





*disable enable

## 35. Data Matrix

**(1) Data Matrix enable/disable**



*** enable**



**disable**

**(2) Read reverse Data Matrix**



*only read Normal code



Only Read reverse code



Read Normal/reverse code

① **Data Matrix Mirror**



***disable**



**enable**

## 36. PDF 417



*** enable**



**disable**

**37. Aztec code**



**enable**



***disable**

**38. Maxi code**



**enable**



***disable**

**39. Hanxin code**



**enable**



***disable**

**40. Brazilian bank code**



**enable**



***disable**

## Appendix 1：numbered bar code

For parameters requiring specific numeric values, scan the appropriately numbered bar code(s).



0



1



2



3



4



5



6



7



8



9

## Appendix 2：CANCEL

To change the selection or cancel an incorrect entry, scan the bar code below.



**Cancel**

## Appendix 3：Code ID

| Code character | Code symbol |
|---|---|
| A | UPC-A, UPC-E, EAN-8, EAN-13 |
| B | Code 39, Code 32 |
| C | Codabar |
| D | Code 128, GS1-128,ISBT 128 |
| E | Code 93 |
| F | Interleaved 2 of 5/ITF, ITF14 |
| G | Industrial 2 of 5, Standard 2 of 5 |
| H | CODE11 |
| J | MSI, MSI/Plessey |
| R | GS1 DataBar-14, GS1 DataBar Limited, GS1 DataBar Expanded, RSS |
| V | Matrix 25 |
| r | PDF417 |
| u | DataMatrix(DM) |
| q | QR |
| a | Aztec Code |
| x | Maxi Code |
| c | HanXin |

## Appendix 4：Character Equivalents

| Scan Value | HEX Value | Keyboard Function Key | Keyboard Ctrl Combination Key |
|---|---|---|---|
| 1000 | 00h | Null | CTRL 2 |
| 1001 | 01h | Keypad Enter | CTRL A |
| 1002 | 02h | Caps lock | CTRL B |
| 1003 | 03h | Right Arrow | CTRL C |
| 1004 | 04h | Up Arrow | CTRL D |
| 1005 | 05h | Null | CTRL E |
| 1006 | 06h | Null | CTRL F |
| 1007 | 07h | Enter | CTRL G |
| 1008 | 08h | Left Arrow | CTRL H |
| 1009 | 09h | Horizontal Tab | CTRL I |
| 1010 | 0Ah | Down Arrow | CTRL J |
| 1011 | 0Bh | Vertical Tab | CTRL K |
| 1012 | 0Ch | Backspace | CTRL L |
| 1013 | 0Dh | Enter | CTRL M |
| 1014 | 0Eh | Insert | CTRL N |
| 1015 | 0Fh | Esc | CTRL O |
| 1016 | 10h | F11 | CTRL P |
| 1017 | 11h | Home | CTRL Q |
| 1018 | 12h | Print Screen | CTRL R |
| 1019 | 13h | Delete | CTRL S |
| 1020 | 14h | tab+shift | CTRL T |
| 1021 | 15h | F12 | CTRL U |

| | | | |
|---|---|---|---|
| 1022 | 16h | F1 | CTRL V |
| 1023 | 17h | F2 | CTRL W |
| 1024 | 18h | F3 | CTRL X |
| 1025 | 19h | F4 | CTRL Y |
| 1026 | 1Ah | F5 | CTRL Z |
| 1027 | 1Bh | F6 | CTRL [ |
| 1028 | 1Ch | F7 | CTRL \ |
| 1029 | 1Dh | F8 | CTRL ] |
| 1030 | 1Eh | F9 | CTRL 6 |
| 1031 | 1Fh | F10 | CTRL - |
| 1032 | 20h | Space | |
| 1033 | 21h | ! | |
| 1034 | 22h | ‘ | |
| 1035 | 23h | # | |
| 1036 | 24h | $ | |
| 1037 | 25h | % | |
| 1038 | 26h | & | |
| 1039 | 27h | ‘ | |
| 1040 | 28h | ( | |
| 1041 | 29h | ) | |
| 1042 | 2Ah | * | |
| 1043 | 2Bh | + | |
| 1044 | 2Ch | , | |
| 1045 | 2Dh | - | |
| 1046 | 2Eh | . | |
| 1047 | 2Fh | / | |
| 1048 | 30h | 0 | |
| 1049 | 31h | 1 | |
| 1050 | 32h | 2 | |
| 1051 | 33h | 3 | |
| 1052 | 34h | 4 | |
| 1053 | 35h | 5 | |
| 1054 | 36h | 6 | |

| | | |
|---|---|---|
| 1055 | 37h | 7 |
| 1056 | 38h | 8 |
| 1057 | 39h | 9 |
| 1058 | 3Ah | : |
| 1059 | 3Bh | ; |
| 1060 | 3Ch | < |
| 1061 | 3Dh | = |
| 1062 | 3Eh | > |
| 1063 | 3Fh | ? |
| 1064 | 40h | @ |
| 1065 | 41h | A |
| 1066 | 42h | B |
| 1067 | 43h | C |
| 1068 | 44h | D |
| 1069 | 45h | E |
| 1070 | 46h | F |
| 1071 | 47h | G |
| 1072 | 48h | H |
| 1073 | 49h | I |
| 1074 | 4Ah | J |
| 1075 | 4Bh | K |
| 1076 | 4Ch | L |
| 1077 | 4Dh | M |
| 1078 | 4Eh | N |
| 1079 | 4Fh | O |
| 1080 | 50h | P |
| 1081 | 51h | Q |
| 1082 | 52h | R |
| 1083 | 53h | S |
| 1084 | 54h | T |
| 1085 | 55h | U |
| 1086 | 56h | V |
| 1087 | 57h | W |

| | | |
|---|---|---|
| 1088 | 58h | X |
| 1089 | 59h | Y |
| 1090 | 5Ah | Z |
| 1091 | 5Bh | [ |
| 1092 | 5Ch | \ |
| 1093 | 5Dh | ] |
| 1094 | 5Eh | ^ |
| 1095 | 5Fh | _ |
| 1096 | 60h | ‘ |
| 1097 | 61h | a |
| 1098 | 62h | b |
| 1099 | 63h | c |
| 1100 | 64h | d |
| 1101 | 65h | e |
| 1102 | 66h | f |
| 1103 | 67h | g |
| 1104 | 68h | h |
| 1105 | 69h | i |
| 1106 | 6Ah | j |
| 1107 | 6Bh | k |
| 1108 | 6Ch | l |
| 1109 | 6Dh | m |
| 1110 | 6Eh | n |
| 1111 | 6Fh | o |
| 1112 | 70h | p |
| 1113 | 71h | q |
| 1114 | 72h | r |
| 1115 | 73h | s |
| 1116 | 74h | t |
| 1117 | 75h | u |
| 1118 | 76h | v |
| 1119 | 77h | w |
| 1120 | 78h | x |

| | | | |
|---|---|---|---|
| 1121 | 79h | y | |
| 1122 | 7Ah | z | |
| 1123 | 7Bh | { | |
| 1124 | 7Ch | \| | |
| 1125 | 7Dh | } | |
| 1126 | 7Eh | ~ | |
| 1127 | 7Fh | | Undefined |

## Appendix 5：Supported bar code types

| 条码类型 | 十六进制值 | 条码类型 | 十六进制值 |
|---|---|---|---|
| Not Applicable | 0x00 | EAN 13 with 5 Supps. | 0x8B |
| Code 39 | 0x01 | EAN 13 | 0x0B |
| Codabar | 0x02 | EAN 13 with 2 Supps. | 0x4B |
| Code 128, Setup128 | 0x03 | EAN 13 with 5 Supps. | 0x8B |
| Discrete 2 of 5 | 0x04 | MSI | 0x0E |
| IATA 2 of 5 | 0x05 | GS1-128 | 0x0F |
| Interleaved 2 of 5 | 0x06 | UPC E1 | 0x10 |
| Code 93 | 0x07 | UPC E1 with 2 Supps. | 0x50 |
| UPC A | 0x08 | UPC E1 with 5 Supps. | 0x90 |
| UPC A with 2 Supps. | 0x48 | Trioptic Code 39 | 0x15 |
| UPC A with 5 Supps. | 0x88 | Bookland EAN | 0x16 |
| UPC E0 | 0x09 | Coupon Code | 0x17 |
| UPC E0 with 2 Supps. | 0x49 | GS1 DataBar-14 | 0x30 |
| UPC E0 with 5 Supps. | 0x89 | GS1 DataBar Limited | 0x31 |
| EAN 8 | 0x0A | GS1 DataBar Expanded | 0x32 |
| EAN 8 with 2 Supps | 0x4A | Code11 | 0x0C |
| EAN 8 with 5 Supps | 0x8A | PDF417 | 0xF0 |
| QR | 0xF1 | Data Matrix(DM) | 0xF2 |
| Aztec Code | 0xF3 | Maxi Code | 0xF4 |
| Veri Code | 0xF5 | Han Xin | 0xF7 |
| AIM128 | 0xA2 | ISSN | 0xA3 |
| PLESSEY | 0xA4 | | |